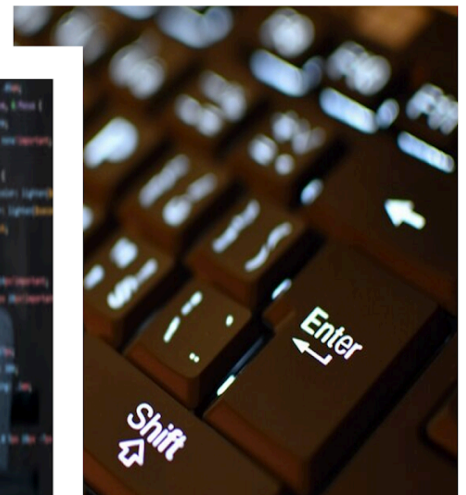


Computer Awareness

STUDY NOTES

Computer Networking



Computer Networking

What is a network?

- Network is defined as a set of devices connected to each other using a physical transmission medium. For Example, A computer network is a group of computers connected with each other to communicate and share information and resources like hardware, data, and software.
- Network, a group of computers and associated devices that are connected by communication facilities.

Need for Networking

- Resource sharing - Through a network , data , s/ w and h/w resources can be shared irrespective of the physical location of the resources and the user.
- Reliability - A file can have its copies on two or more computers of the network.
- Reduced Cost -Sharing resources reduces the cost
- Fast Communication - Information can be exchanged at a very fast speed.

Components of Networks

- Workstation or Nodes : Refers to a computer that are attached to a network
- Server:The master computer is called server. A server is a computer or system that provides resources, data, services, or programs to other computers, known as clients, over a network.
- An individual system can provide resources and use them from another system at the same time. This means that a device could be both a server and a client at the same time.

Different Types of Networks

- There are different categories in which various networks can be classified, according to their size, capabilities and the geographical distance they cover. A network is normally a group of multiple computer systems linked together in some manner so that they can share information and data between them. Various kinds of networks facilitate different services, and have different requirements to work properly.
- Most of the network varieties are known as different types of 'area' networks – this is mainly because of the history of networks, and dates back to the time when computer networks were described by their literal scale. This is no longer the situation due to new technology. Some of the most common types of network user are likely to encounter are detailed here below:

Local Area Network (LAN)

- This is one of the original and very basic type of network, and also one of the simplest. LAN networks group computers together over comparatively small distances, such as within a single building or within a small group of buildings.



Homes usually have LAN networks as well, especially if there are multiple devices in the home. Often they do not contain more than one subnet, if any, and are generally controlled by a single administrator. They are not bound to be linked to the internet to work, even though they can be.

Wide Area Network (WAN)

- This is another kind of the original categories of network, and slightly less typical in nature. WAN networks encapsulates computers together over huge physical distances, remotely connecting them over one huge network and allowing them to communicate even when far apart. The Internet is actually WAN, and connects computers all around the world together.



- LANs connect to WANs, such as the internet, using routers to transfer data and information quickly and securely. WANs are generally too large to be controlled by one administrator, and so generally have collective ownership, or in the case of the internet, is publicly owned.

How do LANs and WANs connect?

- LANs, such as those within a private home, generally have a modem in their residence which is linked to an Internet Service Provider. This provider provides an IP address to the modem, which is an exclusive number that is given to all devices capable of connecting to the internet, including any computers, phones or consoles within the home as well.
- As all devices linked in LAN can interact with each other in absence of the internet, if a device wants to communicate with another which is on another LAN, they are able connect to the internet and transmit information through the WAN. This is implemented through a router, which takes data from devices and routes it

down the fastest virtual path to its end node, passing through a number of gateways on the way.

- Initially a central gateway, which separates the LAN from the WAN, and then others which transmit the data from one to the next, until it is received at its final destination – another device which is being communicated with. This all takes place at a very high speed over modern broadband modems, thanks to efficient and effective protocols and rules being developed to control and manage data.

Other Types of Network

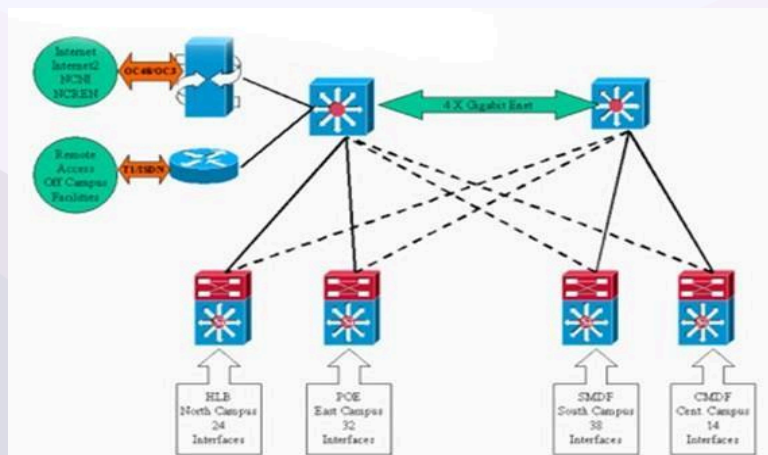
- There are also other types of network user may encounter. Some of these are distinct, but most are casually developed from LAN and WAN networks to have slight variation and adapt to different user needs. These include:

Metropolitan Area Network

- This is a kind of a network which is bigger than a LAN but smaller than a WAN, and incorporates properties of these both. It typically covers a town or city and is controlled by a single person or company, such as a local council or a large company.

Campus Area Network

- This is a network which is bigger as compared to a LAN, but smaller than an MAN. This is very common in areas like a university, large school or small business. It is basically spread over a number of buildings which are reasonably local to each other. It can have an internal Ethernet along with the capability of connecting to the internet.

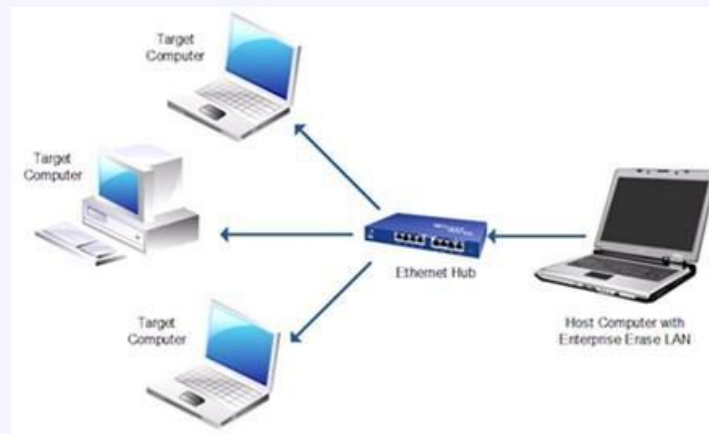


Wireless Local Area Network

- This is a LAN which is implemented with the use of wireless network technology such as Wi-Fi. This kind of network is becoming more popular these days as wireless technology is further developed and is used commonly in the home and in small businesses. In other words, devices do not need to be dependent on physical cables and wires as much and can organise their spaces more effectively.

System Area Network

- The system area network combines computers together on an especially high-speed connection, in a configuration known as a cluster. It means computers which are linked in a group so as to work as a single system, and can be implemented as a result of very high speed computers and new



low cost microprocessors. They are generally used to enhance performance and for cost effectiveness.

Storage Area Network

- This network link servers directly to devices which store large amounts of data without depending on a LAN or WAN network to do so. This can include another type of connection known as

Fibre Channel, a structure similar to Ethernet which manages high-performance disk storage for applications on a variety of professional networks.

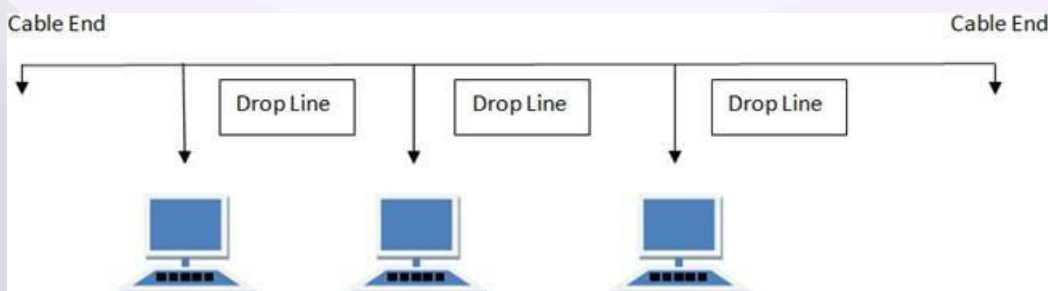
- Normally the two most common network variety user will encounter are **LAN, WAN and WLAN**. This is not a complete and comprehensive collection of all of the types of network, however **most possess similarities** to the ones discussed above as a result of being developed directly based on **older versions such as LAN and WAN**. Different networks are suitable for different needs, and as such, make sure user know his/her way around the types that user are most likely to use.

Types of Network Topology

- The way to arrange or connection patterns of computers/nodes/devices used in the network is known as network topology. There are basically five types of common network topologies:
 1. Bus topology
 2. Star Topology
 3. Ring topology
 4. Mesh Topology
 5. Tree Topology
 6. Hybrid Topology

BUS Topology

Bus topology is a type of network in which each computer and network device is joint to single cable. When it consists exactly two endpoints, then it is called Linear Bus topology.



Features of BUS Topology

1. It sends data only in **one direction**.
2. Every device is linked to a **single cable**.

Advantage of BUS Topology

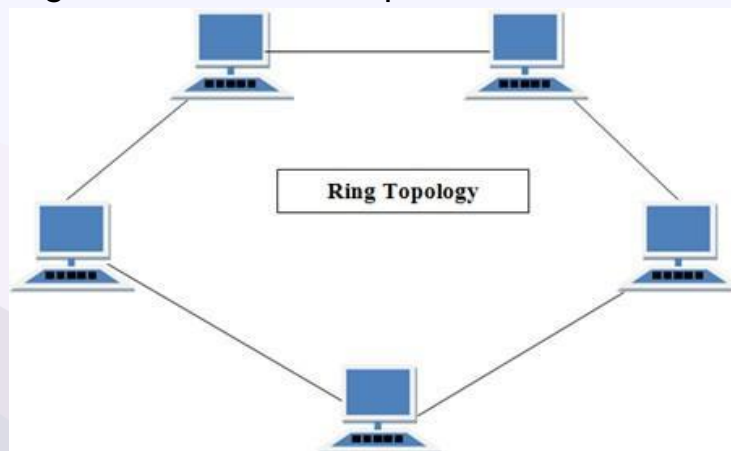
1. Cost effective.
2. Cable requirement is minimal as compared to other topologies.
3. Useful in small networks.
4. Easy to understand.
5. Easy to expand by connecting two cables together.

Disadvantage of BUS Topology

1. Whole network fails if Cables fails then.
2. The performance of the network decreases in case network traffic is heavy or nodes are more or the Cable has a limited length.
3. It is works slower as compared to the ring topology.

RING Topology

- It is named ring topology because it creates a ring as each computer is linked to neighbouring computer, with the last one linked to the first, exactly two neighbours for each computer.



Features of RING Topology

1. Multiple number of repeaters are installed for Ring topology with high number of nodes, because if one node wants to send some data to the last node in the ring topology with more nodes like 100 nodes, then the data will have to travel through 99 nodes in order to reach the 100th node. Hence repeaters are used in the network to prevent data loss.
2. The transmission is in one direction, but it is possible to implement bidirectional transmission by having 2 connections between each Network Node, so called Dual Ring Topology.
3. In case of Dual Ring Topology, two ring networks are created, and data flow is in the opposite direction. Also, if the ring in one direction fails, the second ring can work as a backup, to keep the network working.
4. Data is transmitted in a sequential manner that is bit by bit. Data transmitted, has to go through each node linked in the network, till the final node.

Advantage of RING Topology

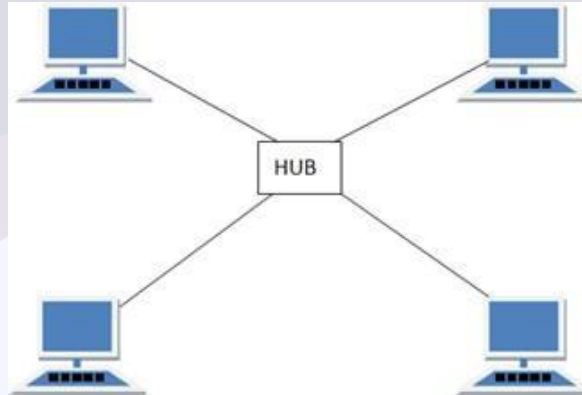
1. Transmitting network is not affected by huge traffic or by addition of more nodes, as only the nodes having tokens are allowed to transmit data.
2. Low cost to install and expand

Disadvantage of RING Topology

1. Troubleshooting is not simple in ring topology.
2. Addition or deletion of the computers interferes the other nodes and network activity.
3. Crashing of one node effects the whole network.

STAR Topology

- In Star Topology all the nodes are linked to a common hub via a cable. This hub is the central node and all others nodes are linked to the central node.



Features of STAR Topology

1. Each node has its own unique connection to the hub.
2. Hub works as a repeater for data transmission.
3. Can be utilized with twisted pair, Optical Fibre or coaxial cable.

Advantage of RING Topology

1. Speedy performance with less number of nodes and low network traffic.
2. Hub can be easily upgraded without hassle.
3. Simpler to troubleshoot.
4. Simpler to setup and modify.
5. Only failed node is affected, rest of the nodes can work without any issues.

Disadvantage of RING Topology

1. Expansive to install.
2. Expensive in usage.
3. If the hub crashes then the entire network is stopped because all linked nodes depend on the hub.
4. Efficiency is dependent on the hub that is it depends on its capacity

MESH Topology

- It is type of a point-to-point to link other nodes or devices. All the network nodes are linked to each other. Mesh consists of $\frac{n(n-1)}{2}$ physical channels to link n number of devices.

- There are two types of methods to transmit data via the Mesh topology, they are :
 1. Routing
 2. Flooding

MESH Topology: Routing

- In routing, the linked computers have a routing logic, as per the network need. Like routing logic to direct the data to transmit to the destination via shortest distance. Or, routing logic which has info related to the broken links, and it neglects those node etc. We can even use routing logic, to re-configure the crashed nodes.

MESH Topology: Routing

- In flooding, the exact data is sent to all the network nodes, hence no routing logic is needed. The network is robust, and data loss is very unlikely. But it results unwanted load over the network.

Types of MESH Topology

1. Partial Mesh Topology: In this type of topology few of the systems are linked in the similar fashion as mesh topology but few nodes are only linked to two or three devices.
2. Full Mesh Topology: Each and every nodes or devices are linked to each other.

Features of MESH Topology

1. Fully linked
2. Robust
3. Not flexible

Advantage of MESH Topology

1. Each connection may carry its unique data load.
2. It is robust.
3. Fault can be diagnosed easily.
4. Facilitates security and privacy.

Disadvantage of MESH Topology

1. Installation and configuration is not simple.
 2. Cabling cost is more.
 3. Bulk wiring is required.
-

TREE Topology

- It has a root node and all other nodes are linked to it creating a hierarchy. It is also called hierarchical topology. It must have minimum three levels to the hierarchy.

Features of STAR Topology

1. Ideal if workstations are situated in groups.
2. Useful in Wide Area Network.

Advantage of RING Topology

1. Extension of bus and star topologies.
2. Expansion of nodes is possible and easy.
3. Easily managed and maintained.
4. Error detection is easily done.

Disadvantage of RING Topology

1. Heavily cabled.
 2. Costly.
 3. If additional nodes are introduced, maintenance is difficult.
 4. Central hub fails, network fails.
-

HYBRID Topology

- It is two different kinds of topologies which is a combination of two or more topologies. For example if in an office in some department ring topology is used and in another department in same place, star topology is used, connecting these topologies will form a Hybrid Topology (ring

topology and star topology).

Features of HYBRID Topology

1. It is a encapsulation of two or topologies
2. Inherits the benefits and disadvantages of the topologies included

Advantage of HYBRID Topology

1. Reliable because Error detecting and trouble shooting is easy.
2. Effective.
3. Scalable as size can be increased easily.
4. Flexible.

Disadvantage of HYBRID Topology

1. Complex in design
2. Costly

Transmission Modes In Computer

Networks

- Transmission mode means transferring of data between two devices. It is also known as communication mode. Buses and networks are designed to allow communication to occur between individual devices that are interconnected.
- There are three types of transmission mode:
 - ◇ Simplex Mode
 - ◇ Half-Duplex Mode
 - ◇ Full-Duplex Model

Simplex Mode

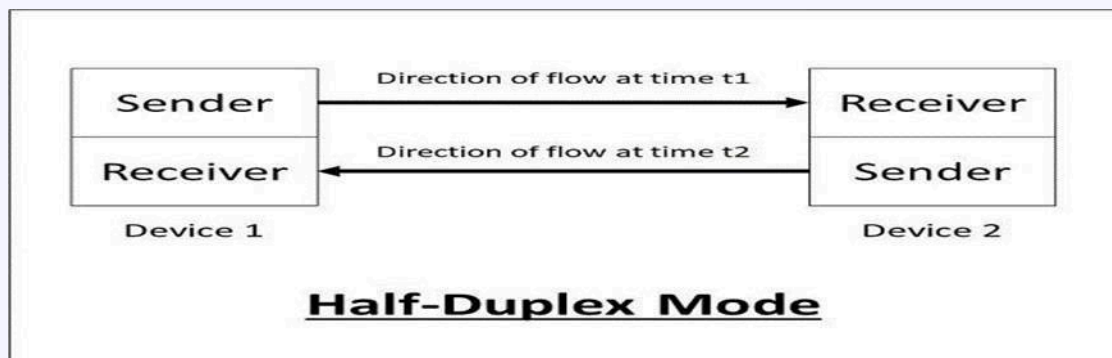
- In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction.

- Example: Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.

Half-Duplex Mode

- In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time. The entire capacity of the channel can be utilized for each direction.

Example: Walkie-talkie in which message is sent one at a time and messages are sent in both directions.



Full-Duplex Mode

In full-duplex mode, both stations can transmit and receive simultaneously. In full_duplex mode, signals going in one direction share the capacity of the link with signals going in other direction, this sharing can occur in two ways:

- Either the link must contain two physically separate transmission paths, one for sending and other for receiving.
- Or the capacity is divided between signals travelling in both directions.

Full-duplex mode is used when communication in both direction is required all the time. The capacity of the channel, however must be divided between the

two directions.

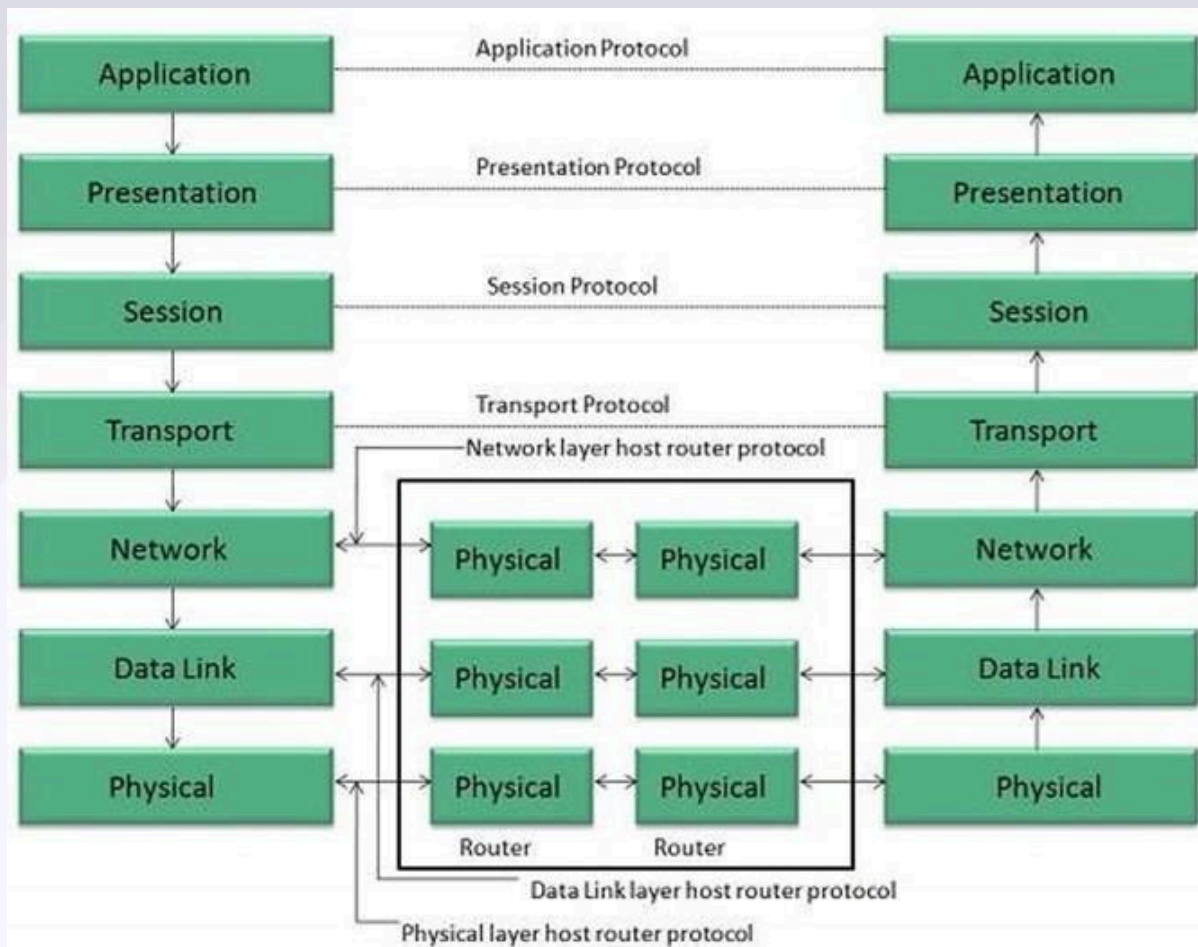
Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

Reference Model

- Reference Model facilitates the ways of standardization which is acceptable worldwide. As people using the computer network are situated over a huge physical range and their network devices might consist heterogeneous architecture. The need to provide connection among heterogeneous devices, we require a standardized model i.e. a reference model, which would facilitate us to communicate using these devices regardless of their architecture.
 - There exists two reference models namely OSI model and TCP/IP reference model, however, the OSI model is basically a hypothetical one but the TCP/IP is absolutely practical model.
-

OSI Model

- OSI is an acronym of Open System Interface. This model type was created by the International organization of Standardization (ISO) and therefore also called ISO- OSI Model.
- The OSI model consists of seven layers as shown in the given diagram. Each layer is responsible for a specific function, however each layer provides services to the layer above.



Physical Layer

- The Physical layer is responsible for the given activities:
 - ◇ Activating, maintaining and deactivating the physical connection.
 - ◇ Defining voltages and data rates needed for transmission.
 - ◇ Converting digital bits into electrical signal.
 - ◇ Deciding whether the connection is simplex, half duplex or full duplex.

Data Link Layer

- The data link layer performs the given functions:
 - ◇ Performs synchronization and error control for the information which is to be transmitted over the physical link.

- ◇ Enables error detection, and adds error detection bits to the data which are to be transmitted.

Network Layer

- Given are the functions of Network Layer:
 - ◇ To route the signals through various channels to the other end.
 - ◇ To act as the network controller by deciding which route data must take.
 - ◇ To divide the outgoing messages into packets and to assemble incoming packets into messages for higher levels.

Transport Layer

- The Transport layer performs the given functions:
 - ◇ It decides if the data transmission must take place on parallel paths or single path.
 - ◇ It performs multiplexing, splitting on the data.
 - ◇ It breaks the data groups into smaller units so that they are handled more efficiently by the network layer.
- The Transport Layer ensures sending of data from one end to other end.

Session Layer

- The Session layer performs the given functions:
 - ◇ Manages the messages and synchronizes conversations between two different applications.
 - ◇ It controls logging on and off, user identification, billing and session management.

Presentation Layer

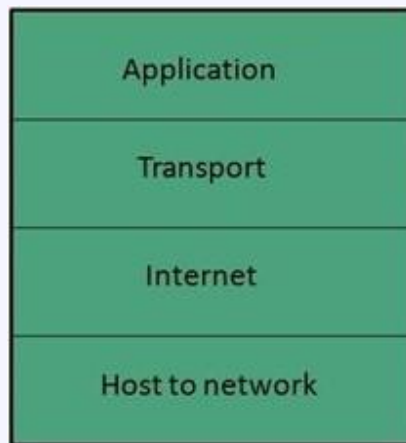
- The Presentation layer performs the given functions:
 - ◇ This layer makes it sure that the information is delivered in such a form that the receiving system will understand and use it.

Application Layer

- The Application layer performs the given functions:
 - ◇ It provides various facilities such as manipulation of information in different ways, retransferring the files of information, distributing the final results etc.
 - ◇ The functions such as LOGIN or password checking are also performed by the application layer.

TCP/IP Model

- TCP/IP model is a practically implemented model, unlike OSI, and is used on the Internet. TCP/IP refers to Transmission Control Protocol and Internet Protocol.
- The TCP/IP model encapsulates the two layers (Physical and Data link layer) into one layer i.e. Host-to-Network layer. The given diagram shows the various layers of TCP/IP model:



Application Layer

- This layer is same as that of the OSI model and performs the given functions:
 - ◇ It provides various facilities such as manipulation of information in different ways, redistributing the files of information, distributing the results etc.
 - ◇ The functions such as LOGIN or password checking are also performed by the application layer.
- Protocols used in this layer: TELNET, HTTP, NNTP, FTP, SMTP, DN are the

protocols employed in this layer

Transport Layer

- It performs the same functions as that of the transport layer present in the OSI model. Here are the key points regarding transport layer:
 - ◇ It uses TCP and UDP protocol for end to end transmission.
 - ◇ TCP is a reliable and connection oriented protocol.
 - ◇ TCP also handles flow control.
 - ◇ The UDP is not reliable and a connection-less protocol also does not perform flow control.
- Protocols utilized in this layer: TCP/IP and UDP protocols are deployed in this layer.

Internet Layer

- The responsibility of this layer is to permit the host to insert packets into the network and then make them travel independently to the destination. However, the sequence of receiving the packet may vary from the sequence they were sent.
- Protocols used: Internet Protocol (IP) is deployed in the Internet layer.

Host - to - Network Layer

- This is at the lowest of the stack among layers in the TCP/IP model. The host has to communicate to the network using some protocol, so that it may transmit IP packets over it. This protocol differs from host to host and network to network.
- Protocols used: ARPANET, SATNET, LAN, packet radio are the protocols which are utilized in this layer.

Network Devices

HUB

- Hub is considered as one of the basic icons of networking devices which is implemented at physical layer and hence connect networking nodes physically together. Hubs are fundamentally utilized in networks that use twisted pair cabling to connect nodes.
- They are created in a way to send the packets to the other appended devices without editing any of the transmitted packets received. They work as pathways to direct electrical signals to travel along. They transmit the information without caring of the fact that the data packet is destined for the device linked or not.

Hub falls in two categories

- **Active Hub:** They are advanced than the passive hubs. They not only facilitate the path for the data signals, but they also regenerate, concentrate and strengthen the signals before transmitting them to their destinations. Active hubs are also called as 'repeaters'.
- **Passive Hub:** They are similar to the point contact for the wires to build in the physical network. They have no work of modifying the signals.

Ethernet Hubs

- It is a device linking various Ethernet devices together and makes them carry on the functions as a single device. They are different in speed in terms of data transfer rate. Ethernet utilizes so called Carrier Sense Multiple Access with Collision Detect (CSMA/CD) to control Media access.



- Ethernet hub connects in half-duplex mode where the possibility of data

collision is inevitable at most of the times.

Switches

- Switches works as the **connection points** for an **Ethernet network**. Just like in hub, devices in switches are linked to them through **twisted pair cabling**. But the variation shows up in the **manner** both the devices, hub and a switch, takes the data. **Hub** works by sending the data to **all the ports on the device** whereas a **switch** transfers it **only to that port which is linked** to the destination device.
- A switch does so by having an in-built learning of the MAC address of the devices linked to it. Since the **sending of data signals are predefined** in a **switch**, therefore the network performance is consequently **effective**. Switches work in **full-duplex** mode where nodes can send and receive data from the switch **simultaneously** unlike in half-duplex mode.
- The transmission bandwidth in **switches is double as compared to the Ethernet hub** transferring around 20Mbps connection into 30Mbps and around 200Mbps connection to become 300Mbps. Performance enhancements are observed in networking with the extensive utilization of switches in the recent days.
- The given method will elucidate further how data transmission takes place via switches:
 - ◇ Cut-through transmission: It allows the packets to be forwarded as soon as they are received. The technique is prompt and quick but the facility of error checking gets overlooked in these kind of packet data transmission.
 - ◇ Store and forward: In this switching technique the entire packet are taken and 'checked' before being transmitted further. The errors are thus eliminated before being propagated further. The drawback of



this process is that error checking spends relatively longer time consequently resulting it a bit slower in processing and delivering.

- ◇ **Fragment Free:** In a fragment free switching environment, a greater part of the packet is examined so that the switch can identify whether the packet has been caught up in a collision. After the collision status is identified, the packet is forwarded.

Bridges

- A bridge is a type of computer networking component that builds the connection with the other bridge networks that are present on the same protocol. It is implemented at the Data Link layer of the OSI Model and links the different networks together and creates communication between them. It combines two local-area networks; two physical LANs into bigger logical LAN or two segments of the already existing LAN that use the same protocol.
- Apart from creating larger networks, bridges are also utilized to segment bigger networks into smaller portions. The bridge performs this by placing itself between the two portions of two physical networks and managing the flow of the data between them.

Types of Bridges

- There are basically **three types** in which bridges can be segmented:
 - ◇ **Transparent Bridge:** As the name signifies, it appears to be transparent for the other devices on the network. The other devices are ignorant of its existence. It only restricts or permits the data according to the MAC address.
 - ◇ **Source Route Bridge:** It received its name from the fact that the way which packet takes via the network is implanted within the packet. It is mainly utilized in Token ring networks.
 - ◇ **Translational Bridge:** The process of conversion happens via Translational Bridge. It transforms the data format of one network to another. For example, conversion of Token ring to Ethernet and vice versa.

Switches Superseding Bridges

- Ethernet switches are gaining trend consecutively as compared to bridges. They are progressing in terms of logical divisions and segments in the networking area. In fact switches are being referred to as multi-port bridges because of their advanced functionality.

Routers

- Routers are network layer components and are particularly categorised as Layer- 3 components of the OSI Model. They work on logical addressing information in the Network header of a packet like IP Addresses.
- Router is utilized to implement bigger complex networks by complex traffic routing. It has the authority to connect dissimilar LANs on the same protocol. It also has the authority to restrict the flow of broadcasts. A router mainly comprises a hardware component or a system of the computer which has multiple network interface and routing software.

Functionality

- When a router receives the data, it analyses the destination address by checking the header of the packet. Once the address is analysed, it finds in its routing table to get know how to connect to the destination and then transmits the packet to the higher hop on the route. The hop might be the last destination or another router.
- Routing tables play a very crucial role in letting the router takes a decision. That is why a routing table needs to be updated and complete. The two methods using which a router can receive data are:
- **Static Routing:** In static routing, the routing information is fed into the routing tables manually. It does not only become a time-taking task but gets prone to errors as well. The manual updating is also required in case of statically configured routers when change in the topology of the network or in the format takes place. Thus static routing is considered for the smallest environments with the least number of one or two routers.
- **Dynamic Routing:** For a larger environment dynamic routing proves to be the practical solution. The process includes use of peculiar routing protocols to establish communication. The purpose of these protocols is to facilitate

the other routers to transmit information about to other routers, so that the other routers can build their own routing tables.

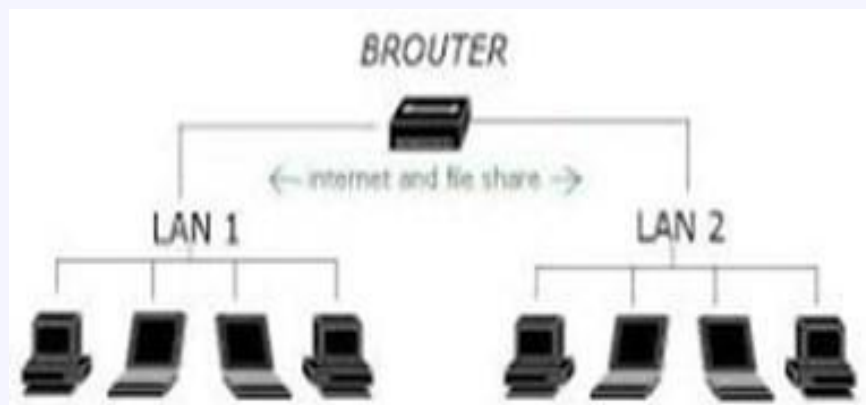
Repeater

- Repeaters are network devices operating at the physical layer of the OSI model that amplify or regenerate an incoming signal before retransmitting it.
- They are incorporated in networks to expand its coverage area. They are also known as signal booster

Brouters

Brouters are the collaboration of bridge and routers. They take up the functionality of the both networking components working as a bridge when forwarding data between networks, and serving as a router when routing information to individual systems.

Brouter works as a filter that permits some data into the local network and redirects unknown packets to the other network.



- These days Brouters are not used because their functionality is embedded into the routers in order to work as bridges as well.

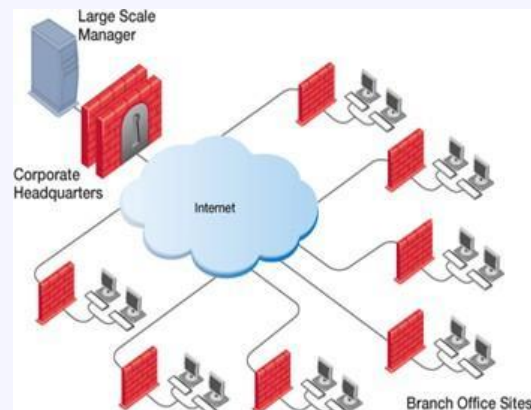
RJ 45 Connector

- RJ45 is a type of cable connector which is mainly used in computer networks. RJ45 is mainly used for ethernet networking which is used to connect different type of devices like a switch, hub, PC, router, firewall to

each other. RJ45 is the most known and popular connector type in the IT world

Gateways

- Gateway is a device which is implemented to combine multiple networks and transmits packets from one packet to the other network. Working as the 'gateway' in middle of different networking systems or computer programs, a gateway is a component which forms a link between them.
- It facilitates computer programs, either on the same device or on different devices to share information across the network via protocols. A router is also a kind of gateway, because it interprets data from one network protocol to another.
- Other components such as bridges translate the data into different forms among two networking systems. Then a software application transforms the data from one format into another. Gateway is a crucial tool to translate the data format, even though the data itself remains unchanged. Gateway might be implemented in some other device to add its functionality into another.



Network Card

- Network cards are sometimes referred to as Network Interface Cards (NICs). These are hardware components that link a computer with the network. They are installed on the mother board. They work for developing a physical connection of the network to the computer. Computer data is transformed into electrical (analog) signals send to the network via Network Interface Cards.
- They are also able to manage some important data-transformation function. These days' network cards are software controlled unlike in older days when drivers were required to configure them. In case if the NIC isn't

packed with the software in the box then the latest drivers or the associated software can be downloaded from the internet also.

Network Protocols

- Network protocols form a language of instructions and translations for communication between the network components. It is mandatory that a networked device should have one or more protocol drivers. Generally, for two computers to interconnect on a network, they must use identical protocols. In some cases, a computer is designed to use more than one protocol. Network protocols such as HTTP, TCP/IP offer provisions on which much of the Internet stands.
- System requirement:
 - ◇ The bus compatibility should be verified on installation of an NIC into the machine. The commonly utilized bus system is Peripheral Component Interconnect (PCI)
 - ◇ Memory I/O addresses and IRQ are required.
 - ◇ Requirement of drivers if not already installed.

Modems

- Modem is a device which transforms the computer-generated digital signals into analog signals to enable their travelling through phone lines. The 'modulator-demodulator' or modem may work as a dial up for LAN or to connect to an ISP.
- Modems can be both external, as in the device which connects to the USB or the serial port of a computer machine, or proprietary components for handheld smart devices and other devices, also as internal, in the form of add-on expansion cards for computer systems and PCMCIA cards for laptop computers.
- Design of a modem differs for both the external and internal modem. In internal modems, IRQ – Interrupt request is utilized to configure the modem in addition to I/ O, which is a memory address. Usually before the installation of built-in modem, integrated serial interfaces are disabled, assigning them the COM2 resources at the same time.

- For external types of a modem, the modem assigns and uses the resources itself. This is especially helpful for the USB port and laptop users as the non-complex and simpler nature of the process renders it far much more useful for daily usage.
- At the time of setup, the second step to make sure the proper working of a modem is the installation of drivers. The modem working efficiency and processing is decided by two factors:
 - Speed of UART –An acronym for Universal Asynchronous Receiver or Transmitter chip (installed on the motherboard to which the modem connection is established)
 - Speed of the modem itself